ANSWER KEY

1. (*b*)	**2.** (*a*)	**3.** (*a*)	**4.** (*a*)	**5.** (*c*)	**6.** (*c*)	**7.** (*b*)	**8.** (*d*)	**9.** (*d*)	**10.** (*b*)
11. (*b*)	**12.** (*c*)	**13.** (*d*)	**14.** (*d*)	**15.** (*b*)	**16.** (*d*)	**17.** (*d*)	**18.** (*d*)	**19.** (*c*)	**20.** (*d*)
21. (*c*)	**22.** (*c*)	**23.** [5]	**24.** [15or75]	**25.** [5]	**26.** [60]	**27.** [30]	**28.** [580]	**29.** [1]	**30.** [15]

Hints and Solutions

1. (*b*) Velocity of the projectile at the highest point $= u\cos\theta$

$$u\cos\theta = \frac{\sqrt{3}u}{2}$$

$\Rightarrow \theta = 30^{\circ}$

Time of flight $T = \dfrac{2u\sin 30^{\circ}}{g} = \dfrac{u}{g}$

2. (*a*) $y = x - \dfrac{x^2}{20}$

For maximum height,

$$\frac{dy}{dx} = 0 \Rightarrow \frac{d}{dx}\left(x - \frac{x^2}{20}\right) = 0$$

$$1 - \frac{2x}{20} = 0$$

$x = 10$

So, $y_{\max} = 10 - \dfrac{100}{20} = 5m$

3. (*a*) At the highest point

Vertical component of the velocity, $v_y = 0$

Horizontal component of the velocity

$v_x = u_x = u\cos\theta$

Gravitational potential energy

$U_g = mgh$, it is maximum at $H_{\max}$.

4. (*a*) Horizontal component of velocity, $v_x = u\cos 30^{\circ}$

$= 40 \times \dfrac{\sqrt{3}}{2} = 20\sqrt{3}$ m/s

Vertical component of velocity, $v_y = u\sin 30^{\circ}$

$= 40 \times \dfrac{1}{2} = 20$ m/s

At $t = 2$s, $v_x = u_x = 20\sqrt{3}$

$v_y = u_y - gt = 20 - 10 \times 2 = 0$

$\therefore\ v = \sqrt{v_x^2 + v_y^2} = 20\sqrt{3}ms^{-1}$

5. (*c*) Maximumheight $H_{\max} = \dfrac{u^2\sin^2\theta}{2g} \Rightarrow H_{\max} \propto \sin^2\theta$

$$\frac{H_1}{H_2} = \frac{\sin^2\theta_1}{\sin^2\theta_2} = \frac{\sin^2 30^{\circ}}{\sin^2 60^{\circ}} = \frac{(1/2)^2}{(\sqrt{3}/2)^2} = \frac{1}{3}$$

6. (*c*) Maximum vertical height is given by

$$H_{\max} = \frac{v^2}{2g} = 136m$$

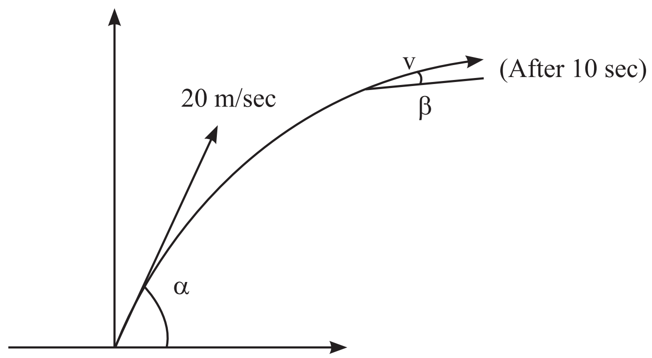
and maximum horizontal range is

$$R_{\max} = \frac{v^2}{g} = 2H_{\max}$$

$= 2(136)$

$\Rightarrow R_{\max} = 272m$

7. (*b*) Through out the motion, the horizontal speed will remain constant, therefore,



Along x-axis,

$u_x = v_x = 20\cos\alpha$ ...(*i*)

Along *y*-axis

$u_y = 20\sin\alpha$,

Applying Kinematic eqn.

$v_y = u_y + a_y t$

$v_y = 20\sin\alpha - 10\times 10$ ...(*ii*)

$$\tan\beta = \frac{v_y}{v_x} = \frac{20\sin\alpha - 100}{20\cos\alpha}$$

$\Rightarrow \tan\beta = \tan\alpha - 5\sec\alpha$

8. (*d*) $y = x - \dfrac{10x^2}{2u^2\left(\frac{1}{2}\right)}$

$$\Rightarrow 10 = 20 - \frac{(10)(400)}{u^2} \Rightarrow u = 20\,\text{m/s}$$

$$t = \frac{T}{\sqrt{2}} = \frac{(2)(20)\times\sin 45^\circ}{\sqrt{2}(10)} = 2\sqrt{s}$$

$$\vec{v} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j} = 10\sqrt{2}\,\hat{i} + \left(10\sqrt{2} - 10(2)\right)\hat{j}$$

Momentum $\vec{p} = M\vec{v} = 100\sqrt{2}\,\hat{i} + \left(100\sqrt{2} - 200\right)\hat{j}$

9. (*d*) $R = \dfrac{V^2(2\sin\theta\cos\theta)}{g}$

$$t = \frac{V\sin\theta}{g} \Rightarrow V = \frac{gt}{\sin\theta}$$

$$\Rightarrow R = \frac{g^2t^2}{\sin^2\theta}\cdot\frac{2\sin\theta\cos\theta}{g}$$

$$\tan\theta = \frac{2gt^2}{R} = \frac{20t^2}{R}$$

$$\cot\theta = \frac{R}{20t^2}$$

10. (*b*) According to question, we can write

$$R = \frac{u^2\sin 2q}{g}$$

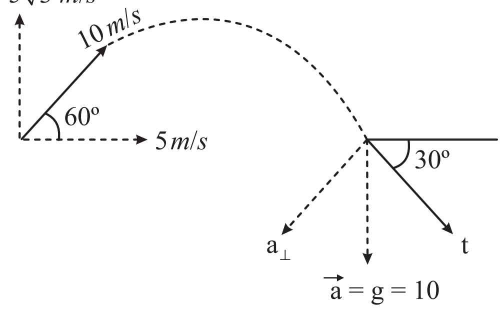
For maximum range, ($R_{\max}$)

$\theta = 45°$.

$$R_{\max} = \frac{u^2}{g} \Rightarrow u^2 = gR_{\max}$$

$$\Rightarrow H_{\max} = \frac{u^2}{2g} = \frac{gR_{\max}}{2g} = \frac{R_{\max}}{2} = \frac{100}{2} = 50\text{ m}$$

11. (*b*) $5\sqrt{3}\ m/s$



At $t = 1$

$u_x = 5,\ u_y = 5\sqrt{3}$

$v_y = 5\sqrt{3} - 10;\ v_x = 5$

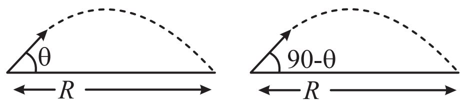
Angle made by velocity vector at $t = 1$ sec.

$$|\tan\alpha| = \left|\frac{v_y}{v_x}\right| = |2 - \sqrt{3}|$$

$\alpha = 15°$

$$\therefore R = \frac{5^2 + \left(10 - 5\sqrt{3}\right)^2}{10\cos\alpha} = \frac{200 - 100\sqrt{3}}{10\times 0.965} = 2.8\text{ m}$$

12. (*c*) Range will be same for time t_1 and t_2, so angles of projection will be 'θ' and '$90° - \theta$'



$$t_1 = \frac{2u\sin\theta}{g}\quad t_2 = \frac{2u\sin(90° - \theta)}{g}$$

$$\text{and } R = \frac{u^2\sin 2\theta}{g}$$

$$t_1t_2 = \frac{4u^2\sin\theta\cos\theta}{g^2} = \frac{2}{g}\left[\frac{2u^2\sin\theta\cos\theta}{g}\right] = \frac{2R}{g}$$

13. (*d*) Equation of trajectory is given as

$y = 2x - 9x^2$ ... (*i*)

Comparing with equation:

$$y = x\tan\theta - \frac{g}{2u^2\cos^2\theta}\cdot x^2 \qquad \text{... }(ii)$$

We get; $\tan\theta = 2$

$$\therefore\ \cos\theta = \frac{1}{\sqrt{5}}$$

Also, $\dfrac{g}{2u^2\cos^2\theta} = 9$

$$\Rightarrow \frac{10}{2\times9\times\left(\frac{1}{\sqrt{5}}\right)^2}=u^2 \quad \Rightarrow u^2=\frac{25}{9} \quad \Rightarrow u=\frac{5}{3}\text{ m/s}$$

14. (*d*) At the maximum height velocity = ucosθ = u cos30°

$$\frac{KE_{initial}}{KE_{top}}=\frac{\frac{1}{2}M(u)^2}{\frac{1}{2}M\left(u\cos30^\circ\right)^2}=\frac{4}{3}$$

15. (*b*) $t=\frac{2\times2\sin15^o}{g\cos30^o}$

$S = 2\cos 15^o \times t - \frac{1}{2}\, g \sin30^o\, t^2$

Put values and solve: $S = 20$ cm

16. (*d*) $\boxed{B}\xrightarrow{V_B}\xleftarrow[V_A]{}\boxed{A}$

Velocity of train *A*, V_A = 90 km/h = – 2.5 m/s

$\because$ 1 km/h $=\frac{5}{18}m/s$

Velocity of train B, V_B = 54 km/h = 15 m/s

Velocity of train B w.r.t. train *A*

$\vec{V}=\vec{V}_B-\vec{V}_A = 15-(-25)$

= 40 m/s

Time of crossing (t) $=\frac{\text{length of train }(l)}{\text{relative velocity}(V)}$

$(8)=\frac{l}{40} \Rightarrow l = 8\times40 = 320$ m

17. (*d*) $X_p(t) = \alpha t + \beta t^2$

$V_p(t)=\frac{dX_P(t)}{dt}=\alpha+2\beta t$ ... (*i*)

$X_Q(t) = ft - t^2$

$V_Q(t)=\frac{dX_Q(t)}{dt}=f-2t$ ... (*ii*)

According to the quastion, (*i*) = (*ii*)

$\alpha + 2\beta t = f - 2t \Rightarrow t(2\beta+2) = f-\alpha$

\ $t=\frac{f-a}{2(b+1)}$

18. (*d*) V_A = 36 km/hr = 10 m/s

V_B = –72 km/hr = –20 m/s

V_{MA} = –1.8 km/hr = – 0.5 m/s

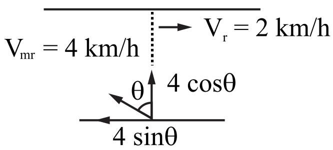
$V_{man,B} = V_{man,A} + V_{A,B}$

= –0.5 + 10 – (–20) = 29.5 m/s

19. (*c*) $\frac{dx}{dt}=y;\frac{dy}{dt}=x$

$\frac{dx}{dy}=\frac{y}{x} \Rightarrow y^2=x^2+c$

20. (*d*)



For swimmer to cross the river straight

$\Rightarrow 4\sin\theta = 2$

$\Rightarrow \sin\theta=\frac{1}{2} \Rightarrow \theta = 30^o$

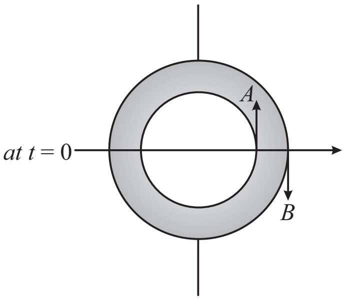
So, angle with direction of river flow = 90° + θ = 120°

21. (*c*) Time taken to reach maximum height

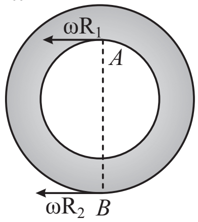
$t=\frac{u\sin\theta}{g} \therefore \frac{u_1\sin\theta_1}{g}=\frac{u_2\sin\theta_2}{g}$

$\Rightarrow u_1\sin30^\circ=u_2\sin45^\circ \quad\Rightarrow \frac{u_1}{u_2}=\frac{1/\sqrt{2}}{1/2}=\frac{\sqrt{2}}{1}$

22. (*c*)



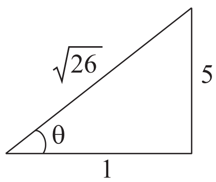
at $t=\frac{\pi}{2\omega}$



$\vec{v}_A-\vec{v}_B=-\omega R_1\hat{i}+\omega.R_2\hat{i} \;=\omega(R_2-R_1)\hat{i}$

23. [5] Given, $u_x=\hat{i},|u_x|=1$

$y=x5(1-x)=x\tan\theta\left(1-\frac{x}{R}\right)$



tanθ = 5, *R* = 1

$$\tan\theta = \frac{|4u_x|}{|4u_y|}$$

$$|4u_y| = |4u_x|\tan\theta = 1 \times 5 = 5 \text{ m/s}$$

24. [15or75] $R = \dfrac{u^2 \sin(2\times 45^\circ)}{g} = \dfrac{u^2}{g}$

$$\frac{R}{2} = \frac{u^2}{2g} = \frac{u^2 \sin 2\theta}{g}$$

$$\Rightarrow \sin 2\theta = \frac{1}{2} \Rightarrow \theta = 15^\circ, 75^\circ$$

25. [5] $u_x = 1$

$$y = 5x(1-x)$$

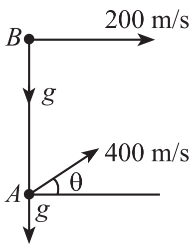
$$\frac{dy}{dt} = 5\frac{dx}{dt} - 10x\frac{dx}{dt}$$

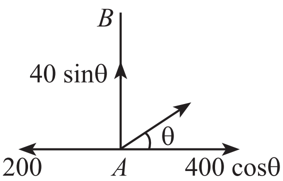
For initial y-component of velocity

$$u_y = \left(\frac{dy}{dt}\right)_{x=0} \Rightarrow 5(1) = 5$$

$$\vec{u}_y = 5\hat{j}$$

26. [60] If A hits B





Then relative velocity perpendicular to the line joining A to B will be zero.

$\Rightarrow 400\cos\theta = 200$

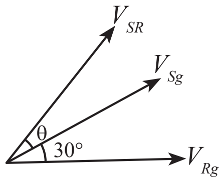
$$\cos\theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

27. [30] Both velocity vector are of same magnitude. Therefore resultant would pass exactly midway through them.

So, $\theta = 30^\circ$

Hence, angle θ with the line AB is 30°



28. [580]

$$X_1 = -3t^2 + 8t + 10$$

$$\vec{v}_1 = (-6t+8)\hat{i} = 2\hat{i}$$

$$Y_2 = 5 - 8t^3$$

$$\vec{v}_2 = -24t^2\hat{j}$$

$$\sqrt{v} = |\vec{v}_2 - \vec{v}_1| = |-24\hat{j} - 2\hat{i}|$$

$$\sqrt{v} = \sqrt{24^2 + 2^2}$$

$$v = 580 \text{ m/s}$$

29. [1] Time of flight is same

$\Rightarrow$ Vertical component of velocity is same

$\Rightarrow H_{\max}$ is same

30. [15] $R_{\max} = \dfrac{u^2 \sin 2(45^\circ)}{g} = \dfrac{u^2}{g}$

$$\frac{R}{2} = \frac{u^2}{2g} = \frac{u^2 \sin 2\theta}{g}$$

$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = 30^\circ, 150^\circ$$

$$\theta = 15^\circ, 75^\circ$$